

BICS 2304

COMPUTER ARCHITECTURE AND ASSEMBLY LANGUAGE

PROJECT TITLE

WEARABLE HEAT STROKE & DEHYDRATION PREVENTION SYSTEM

Submission for Milestone 1: Flowchart, Use Case & Work Distribution

Due: 3 May 2026

SECTION: 04

LECTURER: YASIR MEHMOOD

PREPARED BY:

NO.	NAME	MATRIC NO.	ROLE
1	ADAM FAWWAZ BIN SAZALIZAM	2416969	Member
2	AHMAD FAWWAZ BIN ABDUL KADIR	2418497	Member
3	WAN ABDULLAH MIKHAIEL BIN WAN SHAMSUDDIN	2416229	Member
4	ALFYAN SHAWN BIN JASNI SHIRLIN	2417841	Member

1.0 Project Overview

The Wearable Heat Stroke & Dehydration Prevention System is an integrated IoT solution engineered for high-risk individuals such as outdoor laborers and athletes where it operates in high-heat, high-humidity environments like Malaysia. The system aims to mitigate the risks of exertional heat illness (EHI) by providing real-time physiological monitoring and automated safety interventions.

1.1 Sensor and Actuator Specifications & Logic

Sensor	Technical Description	MIPS Logic
Core Body Temperature Sensor	High-precision thermistor simulation where it measures core temp in Celsius (°C).	Data is polled and compared against a threshold of 38°C in the \$at or \$t registers.
Pulse Rate Sensor	Optical heart rate monitor simulation that measures beats per minute.	Logic triggers a high-stress state if reading exceeds 130 BPM simultaneously with high temp.
Accelerometer	Measures G-force and orientation changes to detect sudden deceleration.	Identifies "Active" vs. "Static" states for the timer and triggers an interrupt-style routine upon zero-motion detection (Fall) .

Table 1. Sensor descriptions

Actuator	Technical Description	MIPS Logic
Haptic Vibration Motor	Provides tactile feedback through pulsed vibration patterns.	Mandatory Rest: Continuous pulse when Heat Stress Index is critical. Hydration: Short double-pulse.
Hydration Reminder LED	High-visibility LED indicator.	Activated after 15 minutes of sustained motion (as tracked by the MIPS software timer).
Emergency GPS Broadcast	Simulated wireless data packet transmission.	Triggered immediately upon Fall Detection ; sends the last recorded coordinates and status code to the Emergency Contact actor.

Table 2. Actuator descriptions

2.0 System Flowchart

The flowchart below illustrates the complete control loop of the embedded controller. After initialisation, the system enters an infinite polling loop that reads all three sensors which are going to evaluate three layered decision conditions in priority order (fall detection first, then heat stress, then hydration timer). Then, it will trigger the appropriate actuator and update the user-facing safety status display before looping back.

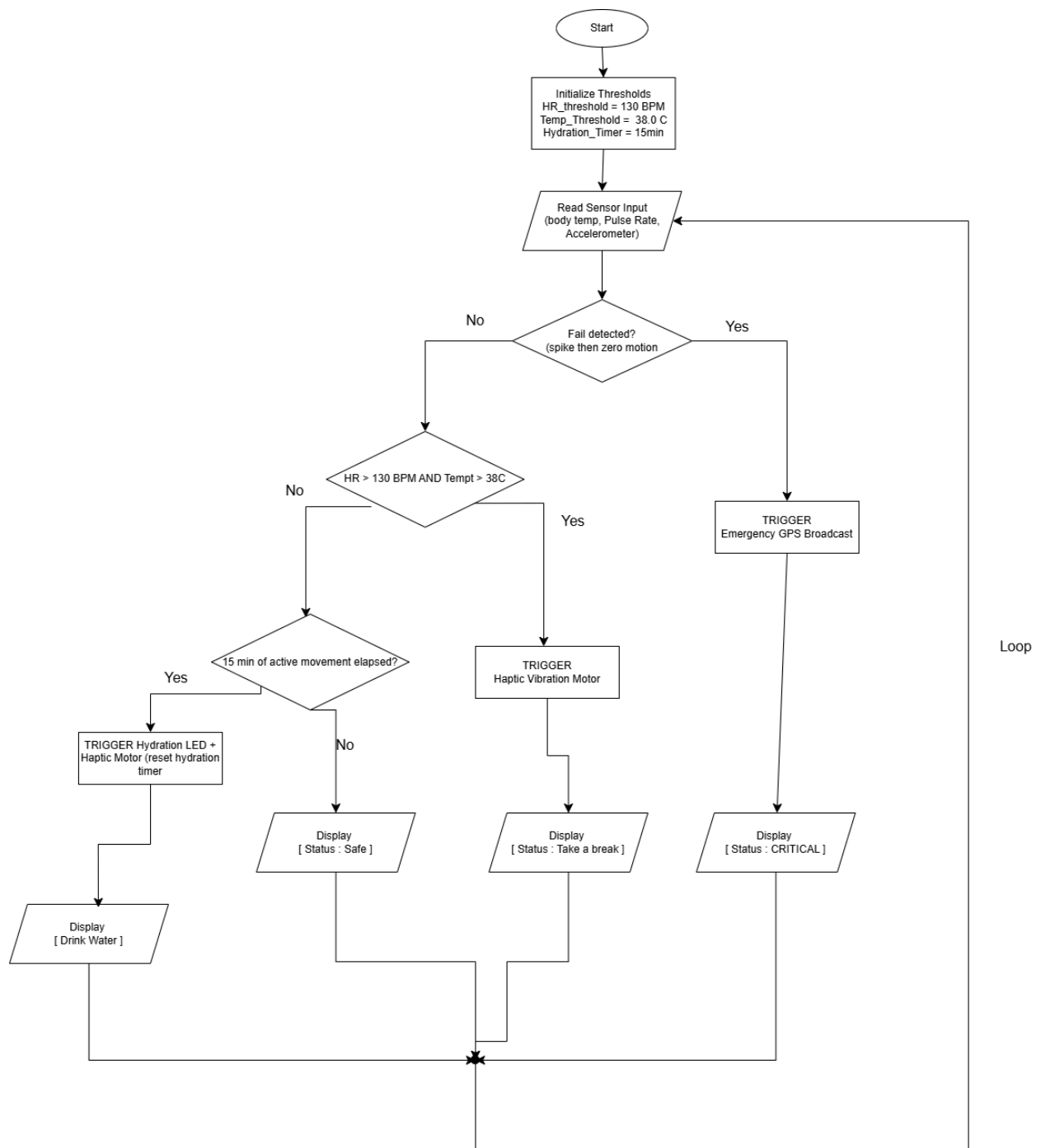


Figure 1. Control flow of the Wearable Heat Stroke & Dehydration Prevention System

3.0 Use Case Diagram

The use case diagram identifies two actors interacting with the system: the Wearer (an outdoor worker or athlete) as the primary actor, and the Emergency Contact as the secondary actor who only becomes involved when a fall is detected. Six core use cases describe the system's behaviour, with three of them sharing a common include relationship to the Monitor Vital Signs use case, and one extend relationship from Send Emergency GPS Alert to Detect Fall.

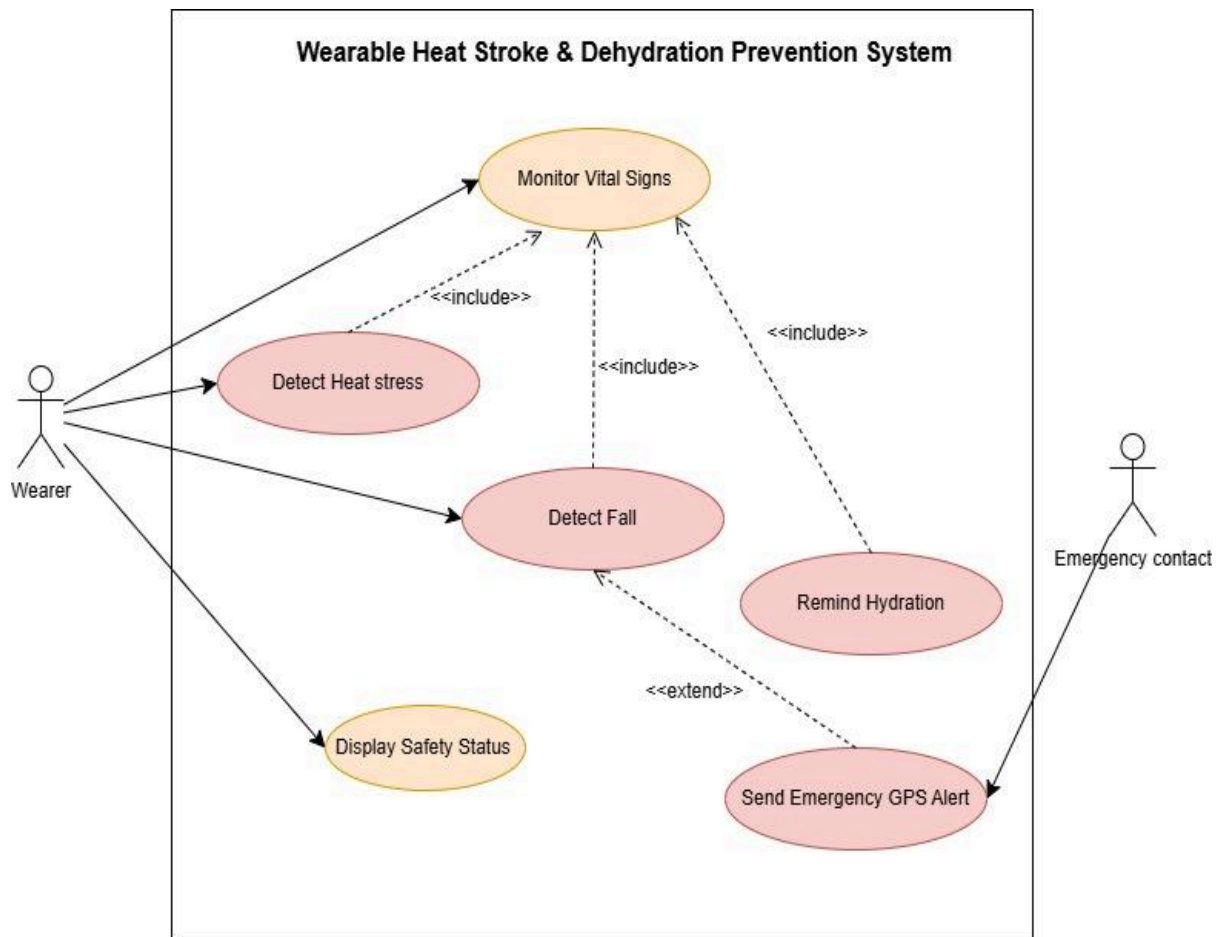


Figure 2. Use case diagram with actors, use cases and stereotype relationships

3.1 Use Case Descriptions

Use Case	Description
Monitor Vital Signs	The system continuously polls the body temperature sensor, pulse rate sensor and accelerometer. The wearer is the data source for all three readings.
Detect Heat Stress	Computes the Heat Stress Index from current body temperature and pulse rate readings. When HR > 130 BPM AND Temp > 38 °C, triggers the haptic motor as a Mandatory Rest alert.
Detect Fall	Identifies a sudden acceleration spike followed by zero motion, indicating that the wearer has collapsed.
Remind Hydration	Tracks elapsed time of sustained active movement. After 15 minutes, activates the Hydration Reminder LED and a brief haptic pulse.
Display Safety Status	Updates the console display with one of four states: [Status: Safe], [Status: Take Break], [Drink Water] or [Status: CRITICAL].
Send Emergency GPS Alert	Extends Detect Fall. When it is triggered, broadcasts an SOS signal with the wearer's last-known status to the Emergency Contact.

Table 3. Use case descriptions

4.0 Work Distribution

The project workload has been divided evenly across all four group members. Each member is assigned one sensor or actuator group, one cross-cutting design responsibility (such as the flowchart, use case diagram, or hardware mapping) and one testing or integration duty. This ensures balanced ownership across the design, prototyping, testing and submission phases described in the project milestones.

NO.	NAME	MATRIC NO.	RESPONSIBILITIES
1	ADAM FAWWAZ BIN SAZALIZAM (Group Leader)	2416969	• Project coordination and milestone tracking • Flowchart design and finalisation • Body Temperature sensor module (MIPS subroutine) • Final integration of all sensor and actuator subroutines
2	AHMAD FAWWAZ BIN ABDUL KADIR	2418497	• Use case diagram and use case descriptions • Pulse Rate sensor module (MIPS subroutine) • Heat Stress Index decision logic and threshold comparison • Unit testing of sensor input modules
3	WAN ABDULLAH MIKHAIEL BIN WAN SHAMSUDDIN	2416229	• Hardware mapping table (simulated to real-world devices) • Accelerometer sensor module (MIPS subroutine) • Fall detection logic and SOS trigger condition • Unit testing of decision and control logic
4	ALFYAN SHAWN BIN JASNI SHIRLIN	2417841	• Display and Safety Status output module • Actuator subroutines: Haptic Motor, Hydration LED, GPS Broadcast • Error handling and input validation across all modules • Final code review and presentation slide preparation

Table 4. Work distribution among group members

4.1 Distribution Rationale

The distribution maps directly to the project's three architectural layers:

- **Sensor input layer:** Members 1, 2 and 3 each own one sensor module.
- **Decision logic layer:** Member 2 owns the Heat Stress Index, Member 3 owns Fall Detection, Member 1 owns the master loop and integration.
- **Actuator and output layer:** Member 4 owns the entire actuator subsystem and the safety status display.

Cross-cutting deliverables (flowchart, use case diagram, hardware mapping table) are split one per member, and each member also owns a testing or quality assurance responsibility to ensure no single person becomes a bottleneck.